

DM-2026-001

SendCUIEmail Development Team
Software Engineering

DATE: January 21, 2026

TO: Project Record

SUBJECT: Cross-Platform PowerShell Support Decision

OFFICE OF PRIMARY RESPONSIBILITY: SendCUIEmail Project

1. **Purpose**

This memorandum documents the decision to implement cross-platform support for SendCUIEmail, enabling the encryption and decryption scripts to run on both Windows PowerShell 5.1+ and PowerShell Core 7+ (macOS and Linux).

2. **Background**

SendCUIEmail was initially designed for Windows 11 environments to encrypt files for secure email transmission of CUI (Controlled Unclassified Information). The tool uses AES-256-CBC encryption with PBKDF2 key derivation, which are standard cryptographic algorithms available in .NET across all platforms.

During development, a requirement emerged to enable developers to test the core encryption/decryption logic on macOS development workstations without requiring a Windows virtual machine. This capability improves development velocity and enables continuous integration testing on non-Windows platforms.

3. **Technical Analysis**

The primary compatibility issue between Windows PowerShell and PowerShell Core is the to plain text conversion. The standard Windows approach uses Marshal methods:

This approach fails on non-Windows platforms because behaves differently. The cross-platform alternative uses the class:

Platform Detection: PowerShell Core exposes the , , and automatic variables. Windows PowerShell (Desktop edition) does not define these, so we detect the platform using:



4. Scope of Changes

The following modifications were made to support cross-platform operation:

- Added helper function that uses platform-appropriate conversion
- Added detection variable
- Updated `in Encrypt.ps1` to use the helper function
- Updated `in Decrypt.ps1` to use the helper function
- Added platform check in `to skip .msg generation on non-Windows`
- Updated script documentation to note cross-platform support

5. Platform-Specific Limitations

The following features remain Windows-only:

- **Outlook .msg Generation:** Requires Outlook COM automation, only available on Windows with Microsoft Outlook installed
- **Batch File Launchers:** The `.bat` files are Windows-specific; macOS/Linux users must invoke PowerShell directly

These limitations are acceptable because:

- The target deployment environment is Windows 11
- macOS/Linux support is primarily for development and testing
- Core encryption/decryption functionality works identically across platforms

6. Security Considerations

No reduction in security posture:

- The cryptographic algorithms (AES-256-CBC, PBKDF2-HMAC-SHA256) are identical across platforms
- The encrypted file format is unchanged and interoperable
- Files encrypted on Windows can be decrypted on macOS/Linux and vice versa
- The approach for SecureString conversion is a documented .NET pattern

7. Decision

APPROVED: Implement cross-platform support as described above.

Rationale:

- Enables faster development iteration without Windows VM dependency
- Supports CI/CD testing on multiple platforms
- No security impact—cryptographic operations are unchanged
- Graceful degradation for Windows-only features (Outlook .msg)
- Maintains full compatibility with existing encrypted files

8. **Testing Requirements**

The following test scenarios must pass on both Windows and macOS/Linux:

- Encrypt single file → Decrypt → Verify contents match
- Encrypt multiple files → Decrypt all → Verify contents match
- Wrong password rejection
- Empty file handling
- Binary files with null bytes
- Cross-platform interoperability (encrypt on macOS, decrypt on Windows)